

Eshwar Pamula

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EDUCATION

The Ohio State University - Honors Scholar

Graduation: December 2025

Bachelor of Science in Mechanical Engineering with Research Distinction (Economics Minor)

WORK EXPERIENCE

Lila Sciences | *Mechanical Engineer I*

January 2026 - Present

- Drove mechanical standardization of the autonomous lab chassis platform; designed mounting interfaces and infrastructure using Solidworks and GD&T tolerance stackups to enable rapid and uniform system deployment
- Designed, prototyped, and installed 10+ custom mounts and labware for scientists in Life and Material Sciences
- Owned mechanical coordination for planar motor transport system; developed installation plans and component readiness schedules using Beckhoff PLC controls and motion validation
- Designed sputtering target swap mechanism with integrated inert gas isolation system; eliminated 2-day post-experiment vacuum recovery by maintaining continuous inert/vacuum environment
- Designed and tested adaptive gripper for 6-DOF robot arm (Franka AMR) for laboratory experiment workflows

Flux Marine | *Powertrain Engineering Intern*

May 2025 - August 2025

- Contributed to the development of the stator for the second-generation in-house electric motor prototype
- Designed and built a shrink-fit jig to reduce assembly process time for the segmented stator core by 35%
- Ran dynamometer testing across varying torque and RPM ranges to validate motor performance metrics

Allegion Inc. | *R&D Mechanical Engineering Intern*

May 2024 - August 2024

- Worked on new product development initiatives (NPI) for new electro-mechanical residential lock set designs
- Designed a dual-CAM interconnect assembly using CREO and validated it through FEA for BHMA standards
- Manufactured early prototypes of the lock system (CNC and PLA & SLA 3D-Printing), to ensure feasibility

RESEARCH EXPERIENCE

Multi-functional Materials and Intelligent Design Lab | *Undergraduate Researcher* *May 2024 - December 2025*

- Awarded a CoE grant and developed machine learning algorithms in Python to reduce computational solving time in topology optimization by having live geometric primitive detection.

Gear and Power Transmission Laboratory | *Undergraduate Research Assistant*

April 2023 - May 2024

- Conducted experimental analysis of Torlon gear properties using Symbrium machines
- Monitored and controlled oil flow, temperature, and torque conditions to assess the performance of gear systems

LEADERSHIP & EXTRACURRICULAR EXPERIENCE

Formula Buckeyes FSAE | *Suspension Engineer*

2022 - August 2024

- Designed and manufactured mounts and hard points using the mill and lathe for the control arms and sway bar
- Contributed to the design, manufacture, and testing of a Front Anti-Roll Bar (T-Bar) for the 2024 racecar
- Played a key role in the team's success, securing 3rd-place in 2023 and 1st-place in 2024 at Michigan FSAE

The Ohio State University: MAE | *Undergraduate Teaching Assistant*

April 2023 - December 2025

- Taught and mentored 500+ students in Statics, Mechanics of Materials, and Machine Elements over 5 semesters
- Awarded Runner Up for the Best Undergraduate Teaching Assistant in the MAE Department (2025)

AWARDS, SKILLS, AND INTERESTS

- **CAD & FEA Software:** SolidWorks, SolidWorks PDM, PTC OnShape, CREO, ANSYS
- **Skills & Tools:** GD&T, 5-Axis CNC, Lathe, 3D Printing, Mill
- **Programming Languages:** MATLAB, Python, Arduino IDE, HTML/CSS, Beckhoff PLC
- **Hobbies:** Poker, Table Tennis, Soccer, 35 mm Film Photography